Project 1 Report: Linear Regression & Logistic Regression

Question 1: Linear Regression

1. Introduction

The objective of this task is to implement Linear Regression from scratch using NumPy and optimize it using Gradient Descent. The model predicts house prices based on 13 features from the Boston Housing dataset.

---

2. Methodology

2.1 Model

The linear regression model is defined as:

[Y = WX + b]

2.2 Loss Function

The loss function used is Mean Squared Error (MSE):

[Loss = 1/n * (sum (y_true - y_pred)^2)

---

3. Implementation Details

- Dataset split into training and testing sets
- Gradient Descent used for optimization
- Parameters updated iteratively using learning rate
- Convergence condition:
  [|| W_{new} - W_{old} ||^2 <= epsilon]

---

4. Learning Rate Experiments

Learning Rate	Observation
1	Too large, unstable, no convergence
0.1	Still unstable, large oscillations
0.01	Good convergence
0.001	Best balance between speed and stability
0.0001	Very slow convergence

Conclusion:
Learning rate **0.001** was selected as it provides stable and efficient convergence.

---

5. Output Graph:

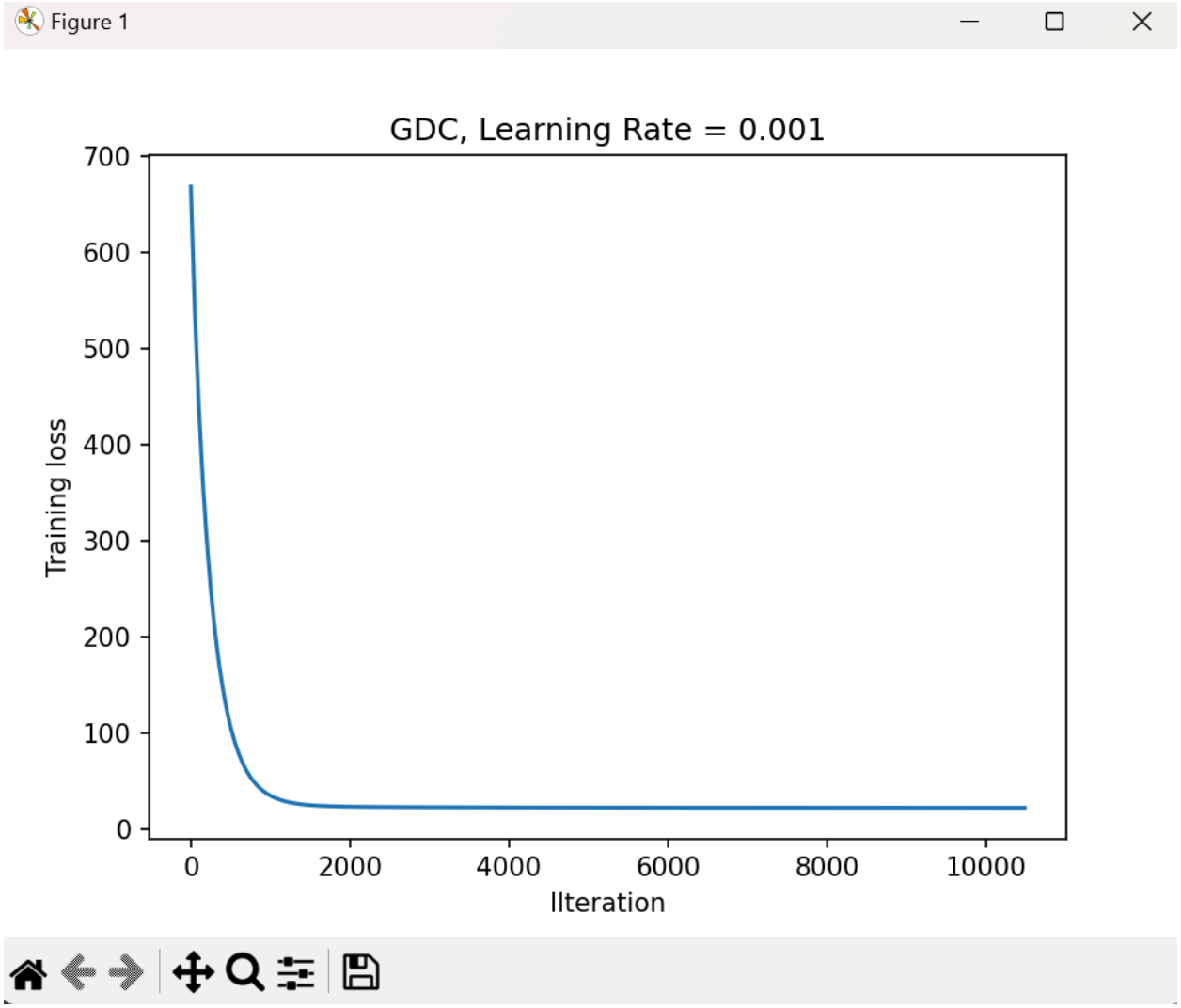


Figure 1
GDC, Learning Rate = 0.001
Training loss
Iteration
700
600
500
400
300
200
100
0
0
2000
4000
6000
8000
10000

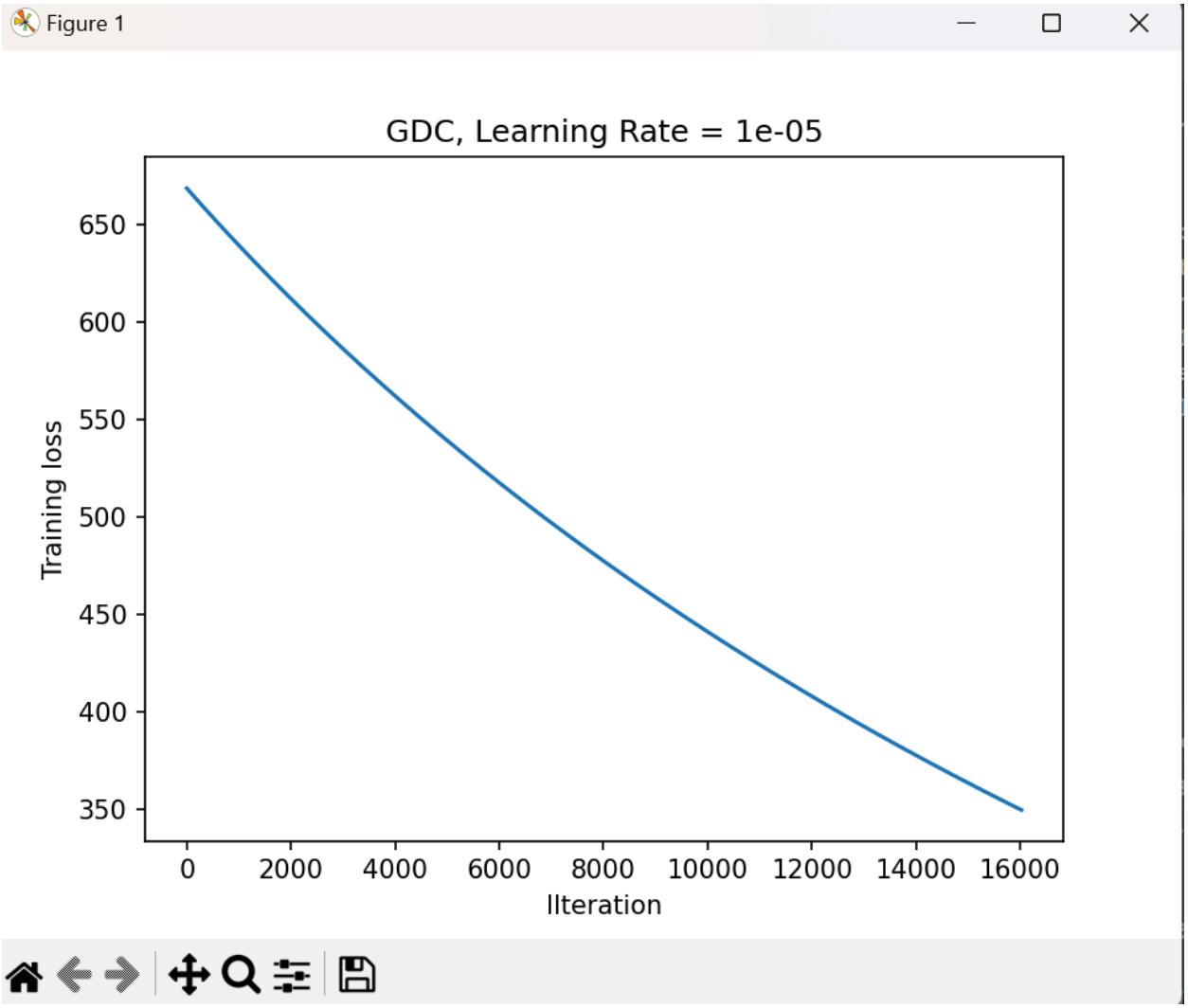
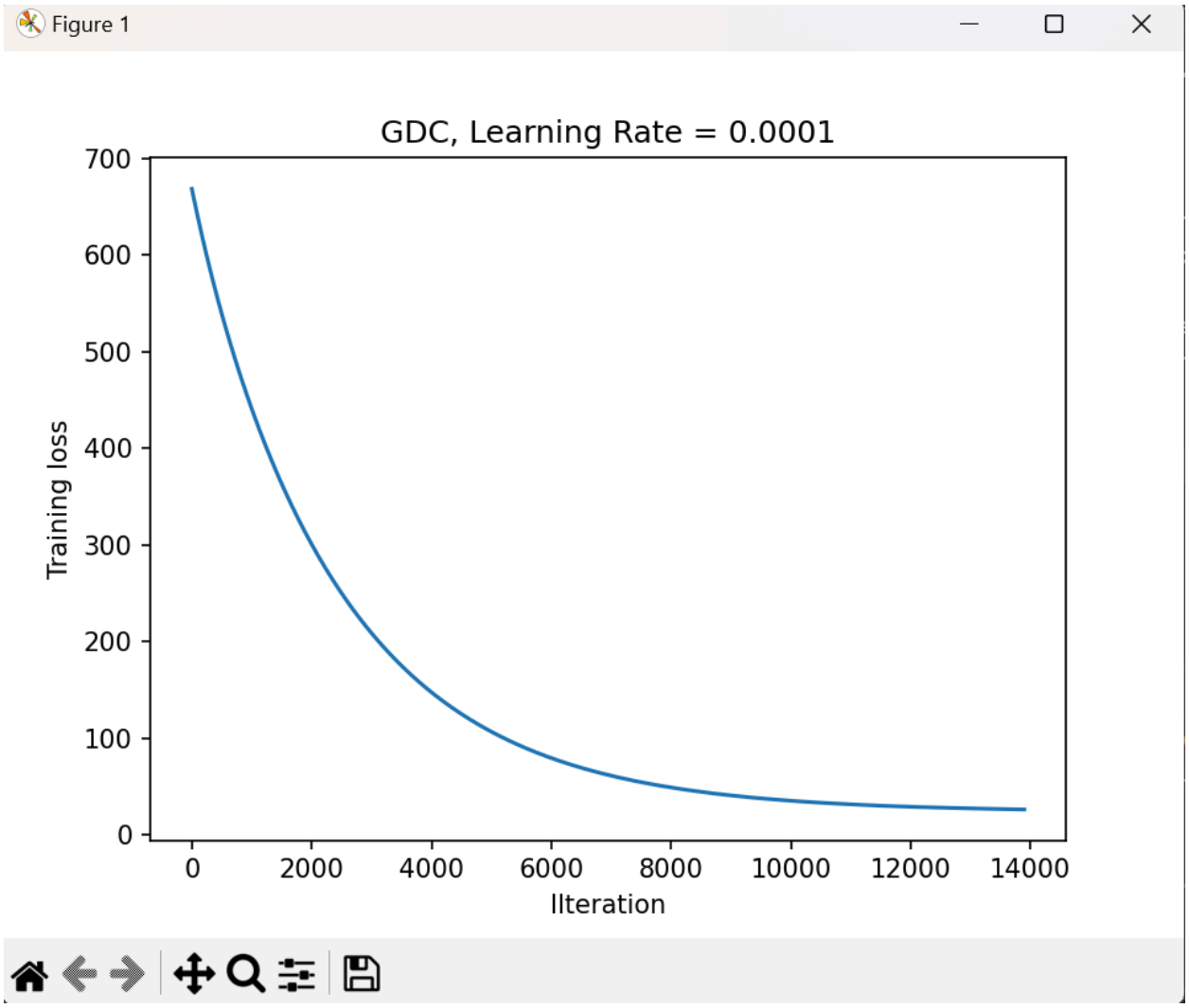


Figure 1
GDC, Learning Rate = 1e-05
Training loss
650
600
550
500
450
400
350
0
2000
4000
6000
8000
10000
12000
14000
16000
Iteration



---

6. Results

- Model successfully trained using gradient descent
- Training loss decreases smoothly over iterations
- Learned weights and bias reported from implementation

---

7. Evaluation

- Mean Squared Error (MSE) computed on test data
- Predictions closely match actual values
- Some failure cases observed where predictions deviate due to outliers

---

8. Comparison with Scikit-learn

- Scikit-learn model performs slightly better due to optimized solvers
- Custom implementation performs comparably well

---

9. Conclusion

The linear regression model successfully learned the relationship between features and house prices. Proper tuning of learning rate was critical for convergence.

Question 2: Logistic Regression

1. Introduction

The goal is to implement Logistic Regression from scratch for binary classification using the Breast Cancer Wisconsin dataset.

---

2. Methodology

2.1 Model

Logistic regression uses the sigmoid function:

[y_hat = 1/(1 + e^(WX + b))]

2.2 Loss Function

Binary Cross-Entropy Loss:

[L = 1/n * sum [y*log(hat{y}) + (1-y)*log(1-hat{y})]]

---

3. Implementation Details

- Sigmoid function implemented
- Gradient descent used for optimization
- Convergence based on parameter updates

---

4. Learning Rate Analysis

Learning Rate	Observation
1	No convergence
0.1	Unstable
0.01	Good convergence
0.001	Best performance
0.0001	Very slow

Final Choice: 0.001

---

5. Output Graph:

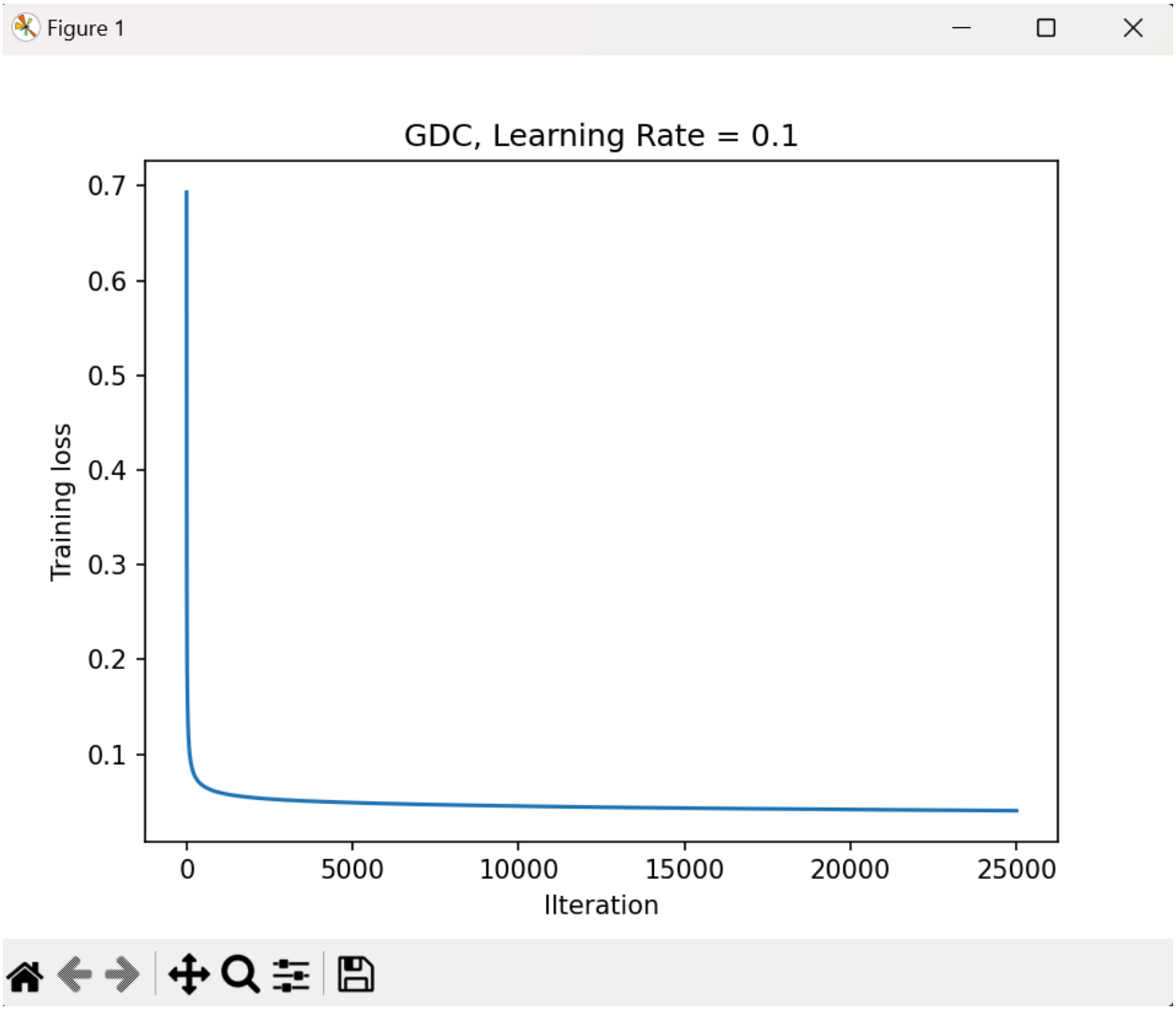


Figure 1
GDC, Learning Rate = 0.1
Training loss
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
5000
10000
15000
20000
25000
Iteration

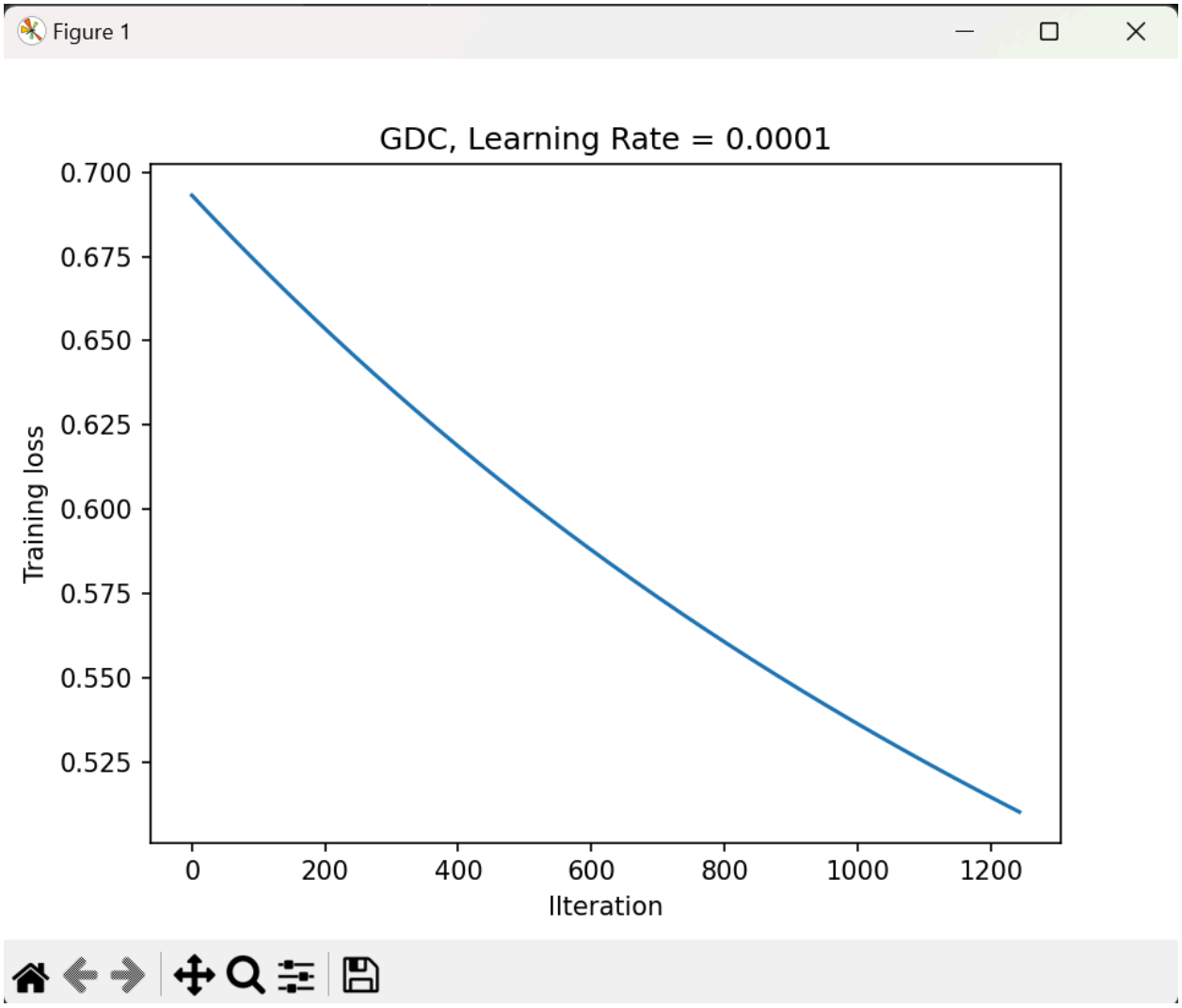
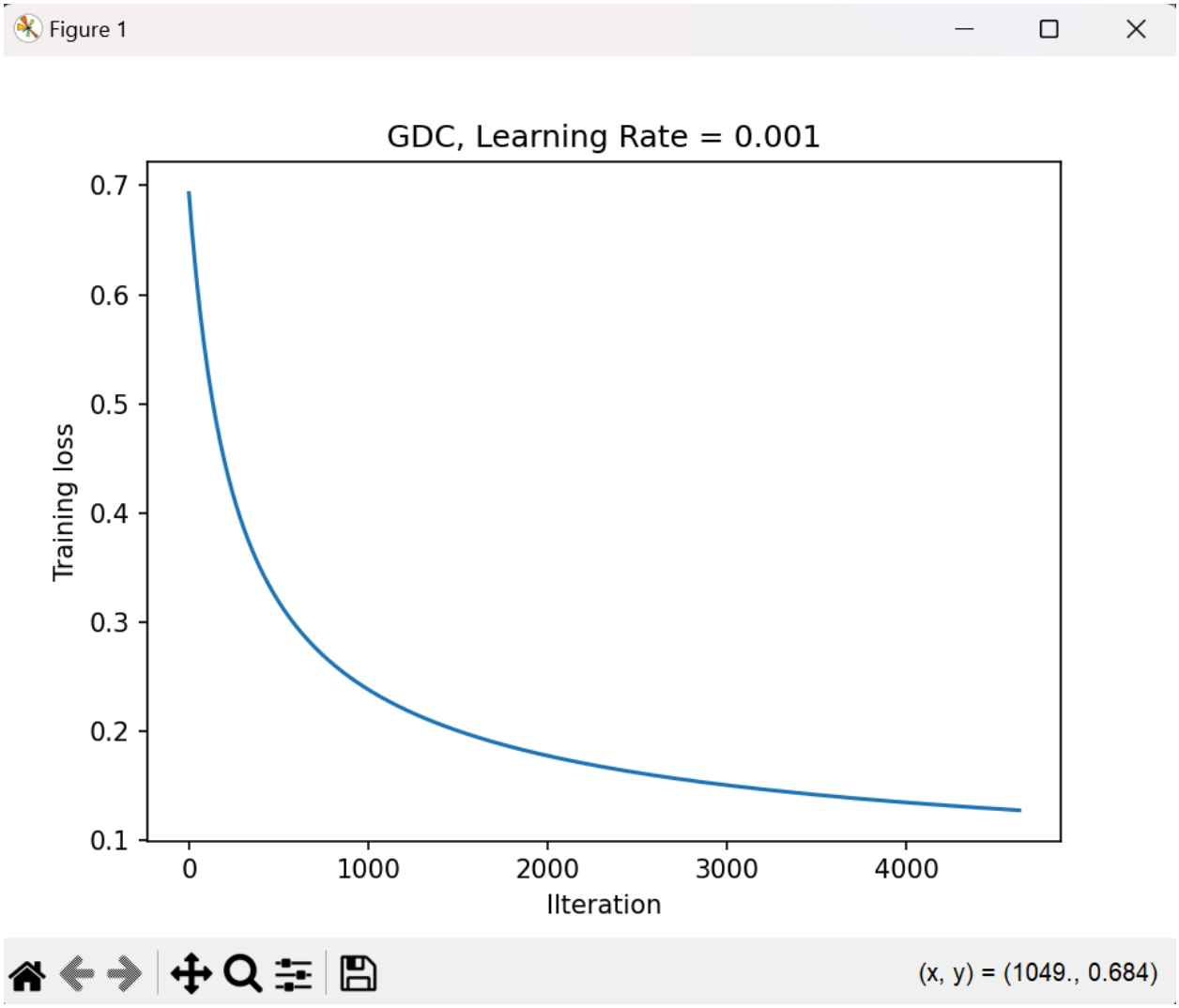


Figure 1
GDC, Learning Rate = 0.0001
0.700
0.675
0.650
0.625
0.600
0.575
0.550
0.525
Training loss
0
200
400
600
800
1000
1200
Iteration



---

6. Results

Training Performance

- Log Loss (Train): 0.1277
- Log Loss (Test): 0.3284

Evaluation Metrics

- Accuracy: 98.24%
- Precision: 96.15%
- Recall: 96.15%
- F1 Score: 96.15%

Confusion Matrix

[[87 , 1]
 1 , 25]]

---

7. K-Fold Cross Validation (k=5)

- Average Accuracy: 97.00%
- Average Precision: 96.62%
- Average Recall: 95.05%
- Average F1 Score: 95.81%

Observation:
Model is stable and generalizes well across folds.

---

8. Comparison with Scikit-learn

Metric	Custom Model	Scikit-learn
Accuracy	98.24%	98.24%
Precision	96.15%	92.85%
Recall	96.15%	100%
F1 Score	96.15%	96.29%

Conclusion:
Both models perform similarly, with slight trade-offs between precision and recall.

---

9. Threshold Tuning

Threshold	Accuracy	Precision	Recall	F1 Score
0.3	0.930	0.765	1.000	0.867
0.5	0.982	0.962	0.962	0.962
0.65	0.974	1.000	0.885	0.939

0.8	0.974	1.000	0.885	0.939

---

10. Analysis of Threshold

- Lower threshold → Higher recall, lower precision
- Higher threshold → Higher precision, lower recall

Insight:
Threshold selection depends on application:

- Medical diagnosis → prioritize Recall
- Spam detection → prioritize Precision

---

11. Conclusion

The logistic regression model performs very well with high accuracy and strong generalization. Gradient descent optimization and threshold tuning significantly impact performance.

Overall Conclusion

- Both Linear and Logistic Regression models were successfully implemented from scratch
- Learning rate tuning is critical for convergence
- Custom implementations perform better and comparable to scikit-learn models
- Logistic regression achieved excellent classification performance

---